

Differential Topology

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Part I

De Rham theory

Chapter 1

De Rham theory

1.1 De Rham theorem

Chapter 2

Hodge theory

elliptic operators

Part II

Cobordism

Chapter 3

Morse theory

3.1 Morse functions

Definition 3.1.1. Let M be a manifold. A *Morse function* is a smooth function $f : M \rightarrow \mathbb{R}$ such that all critical points are nondegenerate.

Proposition 3.1.2. Let M be an embedded submanifold of $\mathbb{R}^n$. For almost every point $p \in \mathbb{R}^n$, the function $f : M \rightarrow \mathbb{R} : x \mapsto \|x - p\|^2$ is Morse.

Proof. Suppose that $p \in \mathbb{R}^n$ makes f be not Morse so that it possesses a degenerate critical point. Note that the notation x can denote not only a point variable on M but also the embedding map $M \hookrightarrow \mathbb{R}^n$. Let $N \subset M \times \mathbb{R}^n$ be the normal bundle of the tangent bundle TM and define a map $\varphi : N \rightarrow \mathbb{R}^n$ such that $\varphi(x, y) = x + y$. We claim that the point $(x, p - x)$ is contained in N and φ is critical at this point if f is degenerate at x .

The differential of f is

$$df_x(v) = 2(x - p) \cdot dx(v) = 2(x - p) \cdot v,$$

so x is critical point if and only if $x - p$ is proportional to $T_x M$.

Let $\{x^i\}_{i=1}^m$ be orthonormal coordinates for M and let $\{e_j\}_{j=1}^{n-m}$ be an orthonormal frame field of N . Define coordinate functions $\{x^i, y^j\}$ on the manifold N by

$$x^i(x, y) := x^i(x), \quad \text{and} \quad y^j(x, y) := y \cdot e_j(x).$$

Then,

$$\left\{ \frac{\partial x}{\partial x^1}, \dots, \frac{\partial x}{\partial x^m}, \frac{\partial y}{\partial y^1}, \dots, \frac{\partial y}{\partial y^{n-m}} \right\}$$

always form an orthonormal basis on $\mathbb{R}^n$ and

Since

$$\frac{\partial \varphi}{\partial x^i} = \frac{\partial x}{\partial x^i} + \frac{\partial y}{\partial x^i} \quad \text{and} \quad \frac{\partial \varphi}{\partial y^j} = \frac{\partial y}{\partial y^j},$$

we have

$$\begin{aligned} \frac{\partial \varphi}{\partial x^i} \cdot \frac{\partial x}{\partial x^k} &= \delta_{ik} - y \cdot \frac{\partial^2 x}{\partial x^i \partial x^k}, & \frac{\partial \varphi}{\partial x^i} \cdot \frac{\partial y}{\partial y^l} &= -y \cdot \frac{\partial^2 y}{\partial x^i \partial y^l}, \\ \frac{\partial \varphi}{\partial y^j} \cdot \frac{\partial x}{\partial x^k} &= 0, & \frac{\partial \varphi}{\partial y^j} \cdot \frac{\partial y}{\partial y^l} &= \delta_{jl}. \end{aligned}$$

To represent $d\varphi(\partial_{x^1}, \dots, \partial_{y^{n-m}})$ with matrix, we can write

$$\begin{pmatrix} \frac{\partial \varphi}{\partial x^i} \\ \frac{\partial \varphi}{\partial y^j} \end{pmatrix} \begin{pmatrix} \frac{\partial x}{\partial x^k} & \frac{\partial y}{\partial y^l} \end{pmatrix} = \begin{pmatrix} \text{id} - y \cdot \frac{\partial^2 x}{\partial x^i \partial x^k} & -y \cdot \frac{\partial^2 y}{\partial x^i \partial y^l} \\ 0 & \text{id} \end{pmatrix}.$$

Then,

$$\frac{\partial^2 f}{\partial x^i \partial x^j} = 2 \left(\text{id} + (x - p) \cdot \frac{\partial^2 x}{\partial x^i \partial x^j} \right)$$

deduces that $d\varphi$ is not surjective at $(x, p - x)$. Therefore, by the Sard theorem, set of such p has measure zero. $\square$

Proposition 3.1.3. *Let M be a manifold. The set of Morse functions is dense in $C^\infty(M)$.*

Proof. Let f be a smooth function on M . Embed M in $\mathbb{R}^{d-1}$ such that $x \mapsto (x_2, \dots, x_d)$. Then, $x \mapsto (f(x), x_2, \dots, x_d)$ gives an embedding into $\mathbb{R}^d$. Define a sequence $\{\varepsilon_n\}_n \subset \mathbb{R}^n$ such that $\varepsilon_n \rightarrow 0$ and the sequence of functions

$$f_n(x) := \frac{\|x + ne_1 + \varepsilon_n\|^2 - n^2}{2n}$$

is Morse, where $\{e_i\}$ denotes the standard basis of $\mathbb{R}^d$. This can be done by the previous proposition. Then,

$$\begin{aligned} f_n(x) &= \frac{(f(x) + n + \varepsilon_n \cdot e_1)^2 + \dots + (x_n + \varepsilon_n \cdot e_d)^2 - n^2}{2n} \\ &= f(x) + \frac{\|x + \varepsilon_n\|^2}{2n} + \varepsilon_n \cdot e_1 \end{aligned}$$

proves that $\|f_n - f\|_{C^k(K)} \rightarrow 0$ on every compact $K \subset M$. $\square$

Theorem 3.1.4 (Morse lemma). *Let p be a nondegenerate critical point of a Morse function f on a manifold M . Then, there exists a local chart (U, φ) of p such that*

$$f \circ \varphi^{-1}(x_1, \dots, x_m) = f(p) - \sum_{i=1}^k x_i^2 + \sum_{i=k+1}^n x_i^2$$

for some k . This chart is called Morse chart.

Proof. $\square$

Corollary 3.1.5. *The critical points of a Morse function are isolated. In particular, on a compact manifold are finitely many critical points of a Morse function.*

3.2 Pseudo-gradients

Definition 3.2.1. Let f be a Morse function on a manifold M . A *pseudo-gradient* adapted to f is a vector field X such that

- (a) $df(X) < 0$ at all noncritical points,
- (b) there is a Morse chart at critical points in which $X = \text{grad } f$, where the metric is induced from the chart.

Proposition 3.2.2. *A pseudo-gradient always exists for any Morse functions.*

Proof. Cover the manifold with charts such that every critical point is contained in a unique chart, which is Morse. For each chart (U, φ) , we can define a vector field on U by

$$X := -d\varphi^{-1}(\text{grad}(f \circ \varphi^{-1})),$$

using the standard metric on $\varphi(U)$. Then, we have

$$df(X) = -\langle \text{grad}(f \circ \varphi^{-1}), \text{grad}(f \circ \varphi^{-1}) \rangle \leq 0,$$

where the equality holds only at critical points. With a partition of unity, the vector fields are combined and easily checked to be pseudogradient. $\square$

Definition 3.2.3. Let p be a critical point of a Morse function f on a manifold M . Denote $\varphi^s : M \rightarrow M$ by the flow of a pseudo-gradient. A *stable manifold* is defined as

$$W^s(p) := \{ x \in M : \lim_{s \rightarrow \infty} \varphi^s(x) = p \},$$

and an *unstable manifold* is defined as

$$W^u(p) := \{ x \in M : \lim_{s \rightarrow -\infty} \varphi^s(x) = p \}.$$

Proposition 3.2.4. *The stable manifolds and unstable manifolds are manifolds. Further, they are diffeomorphic open disks. Moreover, the index of p is equal to*

$$\dim W^u(p) = \text{codim } W^s(p)$$

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Chapter 4

Chapter 5

Part III

Topological quantum field theory

Chapter 6

Chern-Simons theory

6.1 Principal connections

6.1 (Equivariant vector bundles over total spaces). Let $\pi : P \rightarrow M$ be a smooth principal G -bundle for a Lie group G with the Lie algebra $\mathfrak{g}$. Over a G -space P , consider the right G -equivariant vector bundles over P

$$(P \times \mathfrak{g}, R \times \text{Ad}^{-1}), \quad (TP, dR), \quad (T^*P, (R^*)^{-1}), \quad ((P \times \mathfrak{g}) \otimes T^*P, (R \times \text{Ad}^{-1}) \otimes (R^*)^{-1}),$$

with their section spaces

$$\Omega^0(P, \mathfrak{g}) = \Gamma(P \times \mathfrak{g}), \quad \mathfrak{X}(P) = \Gamma(TP), \quad \Omega^1(P) = \Gamma(T^*P), \quad \Omega^1(P, \mathfrak{g}) = \Gamma((P \times \mathfrak{g}) \otimes T^*P),$$

where we use the notations

$$\Omega^k(M, F) := \Gamma((M \times F) \otimes \wedge^k T^*M), \quad \Omega^k(M, E) := \Gamma(E \otimes \wedge^k T^*M),$$

for a vector space F , and a vector bundle E over M . We have another right G -equivariant vector bundle (π^*TM, id) over P , which induces a $C^\infty(M)$ -linear map $\mathfrak{X}(M) \rightarrow \Gamma(\pi^*TM)$. Through this chapter, a *bundle* and a *map* will mean a right G -equivariant vector bundle over P and a right G -equivariant vector bundle map over P respectively, if there would be no confusion on what G and P are.

trace and determinant.

6.2 (Lie algebra-valued forms). Let $\pi : P \rightarrow M$ be a smooth principal G -bundle for a Lie group G with the Lie algebra $\mathfrak{g}$. Consider the universal enveloping algebra $U(\mathfrak{g})$. Then, $\Omega(P, U(\mathfrak{g}))$ is an algebra bundle over P because it is the tensor product of two algebra bundles over P . Concretely, if we denote the wedge product of $\omega_1 \in \Omega^k(P, U(\mathfrak{g}))$ and $\omega_2 \in \Omega^l(P, U(\mathfrak{g}))$ by $\omega_1 \wedge \omega_2 \in \Omega^{k+l}(P, U(\mathfrak{g}))$, then it is computed in the geometric wedge product convention as

$$(\omega_1 \wedge \omega_2)(X_1, \dots, X_{k+l}) := \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \omega_1(X_{\sigma(1)}, \dots, X_{\sigma(k)}) \omega_2(X_{\sigma(k+1)}, \dots, X_{\sigma(k+l)})$$

and we also define the Lie bracket $[\omega_1, \omega_2] \in \Omega^{k+l}(P, U(\mathfrak{g}))$ such that

$$[\omega_1, \omega_2](X_1, \dots, X_{k+l}) := \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) [\omega_1(X_{\sigma(1)}, \dots, X_{\sigma(k)}), \omega_2(X_{\sigma(k+1)}, \dots, X_{\sigma(k+l)})]$$

and

$$\begin{aligned} d\omega(X_0, X_1, \dots, X_k) &:= \frac{1}{k+1} \sum_{0 \leq i \leq k} (-1)^i X_i(\omega(X_0, \dots, \hat{X}_i, \dots, X_k)) \\ &+ \frac{1}{k+1} \sum_{0 \leq i < j \leq k} (-1)^{i+j} \omega([X_i, X_j], X_0, \dots, \hat{X}_i, \dots, \hat{X}_j, \dots, X_k). \end{aligned}$$

For example, let $\omega \in \Omega^1(P, \mathfrak{g})$. We can embed $\Omega(P, \mathfrak{g}) \subset \Omega(P, U(\mathfrak{g}))$ to do computations. Then,

$$(\omega \wedge \omega)(X, Y) = \omega(X)\omega(Y) - \omega(Y)\omega(X) = [\omega(X), \omega(Y)]$$

and

$$[\omega, \omega](X, Y) = [\omega(X), \omega(Y)] - [\omega(Y), \omega(X)] = 2[\omega(X), \omega(Y)]$$

imply that $\omega \wedge \omega = \frac{1}{2}[\omega, \omega] \in \Omega^2(P, \mathfrak{g})$. We can also compute the differential by

$$d\omega(X, Y) = \frac{1}{2}(X(\omega(Y)) - Y(\omega(X))) + \frac{1}{2}\omega([X, Y]).$$

The coefficient conventions are not so important that coefficients are eventually cancelled when we write an equation of forms.

6.3 (Ehresmann connections). Let $\pi : P \rightarrow M$ be a smooth fiber bundle.

parallel transport

6.4 (Connection forms). Let $\pi : P \rightarrow M$ be a smooth principal G -bundle for a Lie group G with the Lie algebra $\mathfrak{g}$. Keep in mind that the vertical bundle $VP := \ker(TP \rightarrow \pi^*TM)$ over P , is always canonically trivialized into $P \times \mathfrak{g}$ as the image of the injective map

$$P \times \mathfrak{g} \rightarrow TP : (p, A) \mapsto \left. \frac{d}{dt} p e^{tA} \right|_{t=0},$$

the composition of $P \times \mathfrak{g} \rightarrow TP \times TG : (p, A) \mapsto ((p, 0), (e, A))$ followed by the differential $T(P \times G) \rightarrow TP$ of the principal action $P \times G \rightarrow P$. Now consider the exact sequence of bundles

$$0 \rightarrow P \times \mathfrak{g} \rightarrow TP \rightarrow \pi^*TM \rightarrow 0,$$

where G acts on π^*TM trivially.

An *Ehresmann connection* on P is a choice of splitting section map $\pi^*TM \rightarrow TP$. It is equivalent to a Whitney summand HP of VP in TP over P . The corresponding $C^\infty(M)$ -linear map $\mathfrak{X}(M) \rightarrow \mathfrak{X}(P)$ is called the *horizontal lift*. An Ehresmann connection can be defined for not only principal bundles but also general smooth fiber bundles without G -action.

A *connection form*, on the other hand, is a retraction map $TP \rightarrow VP$. In other words, it is defined as a $C^\infty(P)$ -linear map $\omega : \mathfrak{X}(P) \rightarrow \Omega^0(P, \mathfrak{g})$ or a Lie algebra-valued 1-form $\omega \in \Omega^1(P, \mathfrak{g})$ that is

- (i) right G -invariant in the sense that $\omega(dR_g X) = \text{ad}_g^{-1}(\omega(X)) \in \Omega^0(P, \mathfrak{g})$ for all $X \in \mathfrak{X}(P)$,
- (ii) vertical in the sense that $\omega(A^\#) = A$ for constant sections $A \in \mathfrak{g}$.

6.5 (Curvature form). Let $\pi : P \rightarrow M$ be a smooth principal G -bundle for a Lie group G with the Lie algebra $\mathfrak{g}$. The *curvature form* of the connection form $\omega \in \Omega^1(P, \mathfrak{g})$ can be defined either the covariant derivative of ω or the Cartan structural equation

$$\Omega = d\omega + \omega \wedge \omega = d\omega + \frac{1}{2}[\omega, \omega] \in \Omega^2(P, \mathfrak{g}).$$

The curvature form is horizontal in the sense that for every vertical vector field $X \in \mathfrak{X}(P)$ we have $\iota_X \Omega = 0 \in \Omega^1(P, \mathfrak{g})$.

Proof. Let $X \in \mathfrak{X}(P)$ be a vertical vector field. Then, □

6.6 (Exterior covariant derivatives). Let $\pi : P \rightarrow M$ be a smooth principal G -bundle for a Lie group G with the Lie algebra $\mathfrak{g}$. For a faithful left smooth representation F of G , we have a trivial bundle $P \times F$, where the action is given such that $(p, f)g = (pg, g^{-1}f)$, and the associated vector bundle $E := P \times_G F$ over M .

We say a F -valued form $\omega \in \Omega^k(P, F)$ is *horizontal* if $\iota_X \omega = 0$ for vertical $X \in \mathfrak{X}(P)$. If we denote by $\Omega_h^k(P, F)^G$ the right G -invariant horizontal F -valued k -forms on P , then we have

$$\Omega^k(M, E) \cong \Omega_h^k(P, F)^G.$$

It is because

The exterior derivative $d : \Omega^k(P, F)^G \rightarrow \Omega^{k+1}(P, F)^G$ does not preserve horizontality in general. For a connection form $\omega \in \Omega^1(P, \mathfrak{g})^G$, we can define the *exterior covariant derivative* $\nabla : \Omega^0(M, E) \rightarrow \Omega^1(M, E)$ is defined such that we have a commutative diagram

$$\begin{array}{ccc} \Omega^k(M, E) & \xrightarrow{\nabla} & \Omega^{k+1}(M, E) \\ \parallel & & \parallel \\ \Omega_h^k(P, F)^G & \xrightarrow{d_\omega} & \Omega_h^{k+1}(P, F)^G, \end{array}$$

where d_ω is defined by

$$d_\omega \psi(X_0, \dots, X_k) := d\psi(HX_0^*, \dots, HX_k^*),$$

where $HX_i = X_i - \omega(X_i) \in \mathfrak{X}(P)$ are horizontal components of $X_i \in \mathfrak{X}(P)$. The most important case is $k = 0$.

$$\nabla_X s =$$

6.7 (Maurer-Cartan form). Note that G is itself a principal G -bundle over a point. There is a natural connection form $\omega_G \in \Omega^1(G, \mathfrak{g})$ called the *Maurer-Cartan form*, defined by $TG \rightarrow G \times \mathfrak{g} : (g, X) \mapsto (g, dL_g^{-1}X)$ over a principal right G -homogeneous space G , where G acts $G \times \mathfrak{g}$ from on by the product of the right action and the inverse adjoint.

$$(g, X)h = dR_h(g, X) = (R_h g, dR_h X) \mapsto (R_h g, dL_{gh}^{-1} dR_h X) = (R_h g, dL_h^{-1} dR_h dL_g^{-1} X) = (R_h g, \text{Ad}_h^{-1} dL_g^{-1} X) = (g, dL_g^{-1} X)h$$

The connection on the trivial principal G -bundle $M \times G \rightarrow M$ induced from the Maurer-Cartan connection is also called the Maurer-Cartan connection.

The Maurer-Cartan equation means that the Maurer-Cartan form is flat. It is intuitively clear in some sense that a connection on a point need to be trivially flat.

6.8 (Local expression of principal connections). Let $\pi : P \rightarrow M$ be a smooth principal G -bundle for a Lie group G with the Lie algebra $\mathfrak{g}$. Then, we can fix $\{\varphi_\alpha\}$ be a local trivialization of π such that $\varphi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times G$ is right G -equivariant over U_α , and the local representation of a connection is computed by the composition of right G -equivariant vector bundle maps over $U_\alpha \times G$ or $\pi^{-1}(U_\alpha)$

$$TU_\alpha \times TG \xrightarrow{(d\varphi_\alpha)^{-1}} T(\pi^{-1}(U_\alpha)) \xrightarrow{\omega} \pi^{-1}(U_\alpha) \times \mathfrak{g} \xrightarrow{\varphi_\alpha \times \text{id}} U_\alpha \times G \times \mathfrak{g}$$

$$((x, v), (g, X)) \mapsto (x, g, dL_g^{-1}X + A(x)v)?$$

where $A : TU_\alpha \rightarrow U_\alpha \times \mathfrak{g}$ is over U_α and $A(x) : T_x U_\alpha \rightarrow \mathfrak{g}$ for $x \in U_\alpha$?

If $h(x) = e^{\varepsilon H(x)}$ is a local gauge action for $H : U_\alpha \rightarrow \mathfrak{g}$, then

$$A(x) \rightarrow h^{-1}(x)A(x)h(x) = (1 - \varepsilon H(x))A(x)(1 + \varepsilon H(x)) = A(x) - \varepsilon[H(x), A(x)]$$

as $\varepsilon \rightarrow 0$. Since $h'(x) = \varepsilon h(x)H'(x)$, total derivative? Locally P is $M \times G$, where M is the Minkowski, G is the gauge group, so gauge fields get Poincaré action and the gauge action.

- (a) The above local representation is a right G -equivariant affine bundle map.
- (b) The above local representation is injective and the image is exactly the set of connection forms on the trivial principal G -bundle $\pi^{-1}(U_\alpha)$ over U_α .
- (c) What is the meaning of the expression $\nabla = d + A$?
- (d) Local expression of $\Omega = d\omega + \omega \wedge \omega$ is given by $F = dA + A \wedge A$?
- (e) What is the meaning of the expression $dA \wedge A + \frac{2}{3}A \wedge A \wedge A$ and how can we write it in terms of principal connections.

Proof. (a) We interpret $A \in \Omega^1(U_\alpha, \mathfrak{g})$ as $A : \mathfrak{X}(U_\alpha) \rightarrow \Omega^0(U_\alpha, \mathfrak{g})$ so that for every $X \in \mathfrak{X}(U_\alpha)$,

(c)

Let $L \rightarrow M$ be a trivial line bundle.

Let $s \in \Omega^0(M, L)$, so that $ds : \Omega^1(M, L)$ or

$$ds : \mathfrak{X}(M) \rightarrow \Omega^0(M, L) : X \mapsto X^\mu \partial_\mu s.$$

Let $A \in \Omega^1(M, \mathfrak{u}(1))$ or

$$A : \mathfrak{X}(M) \rightarrow \Omega^0(M, \mathfrak{u}(1)) : X \mapsto X^\mu A_\mu.$$

We have $\nabla = d + A$ in the sense that

$$\nabla_X s = (\nabla s)(X) = ((d + A)s)(X) = X^\mu \partial_\mu s + X^\mu A_\mu s$$

(d)

□

$$S := \frac{k}{4\pi} \int_M \text{tr} \left(dA \wedge A + \frac{2}{3} A \wedge A \wedge A \right)$$

In this case, the field equation is $F = 0$?

6.2 Chern-Weil theory

$$(\text{Sym}^n \mathfrak{g}^*)^G \cong (\text{Sym}^n \mathfrak{t}^*)^W \cong H^{2n}(BG, \mathbb{R}).$$

Given a principal G -bundle $P \rightarrow X$, this isomorphism induces a ring homomorphism

$$\text{CW} : (\text{Sym}^* \mathfrak{g}^*)^G \rightarrow H^{\text{ev}}(X, \mathbb{R}),$$

called the *Chern-Weil homomorphism*. In fact, when X is a smooth manifold, then there is a direct construction of the Chern-Weil homomorphism using connections. Choose any connection ω on P , and let Ω be the curvature.

$$(\mathrm{Sym}^n \mathfrak{g}^*)^G \otimes \Omega^{2n}(P, \mathfrak{g}^{\otimes n}) \rightarrow \Omega^{2n}(P)$$

6.3 Chern-Simons invariants

6.4 Differential cohomology

Chapter 7

Chapter 8

Part IV

Symplectic geometry

Chapter 9

Chapter 10

Chapter 11